17 – MCP (Model Context Protocol)

Definition: MCP (Model Context Protocol) is an open standard that connects AI apps to programs and data in a uniform way – a “USB-C plug” between AI and services.

> **Remember:** One plug for everything: MCP connects the AI to your calendar, files, and more – without a special solution for every single program.

Key points

- **The problem without a standard:** Every AI app needs its own connection for every program (calendar, files, databases). That’s laborious and error-prone.
- **Building blocks:**
  - **Client:** the AI app that uses the connections
  - **Server:** the connection to a specific program or system
  - **Tools/resources/prompts:** what the server offers – actions, data, templates
- **Examples:** calendar (read/create appointments), file system (search folders), databases, project management, knowledge stores.
- **Benefits:** Build once, use everywhere. Fewer one-off solutions, interchangeable building blocks.
- **Servers over hype:** The standard makes connecting easy – but whether a connection is sensible and safe is still your decision.
- **Security:** Only trusted servers, keep permissions minimal (read before write), review permissions regularly. Combined with third-party content, a server can also become an attack route (topic 12).
- **The term isn’t the point:** Some providers build similar things under different names. The principle matters, not the acronym.

Example

You want your AI app to know your work calendar:

- **Without MCP:** You copy your appointments into the chat every morning. Clunky, and currently manual only.
- **With MCP:** The app connects through a calendar server. You ask: “What appointments do I have on Thursday? When does the class reunion fit?” The AI reads real data and, with your approval, can create an appointment.

The key point: the server only needs **read permission** for the first task. Turn on write access (creating appointments) only when you really need it – and confirm every action yourself.

Do / Don’t

Do	Don’t
Use servers only from trusted sources	Install just any server from the web
Read permissions before write permissions	Grant full access to email, files, and calendar at once
Review permissions regularly	Grant approvals once and forget them

Safety note

An MCP server can become dangerous in this combination: third-party content (emails, documents) contains hidden instructions (topic 12), and the server holds the permissions to follow them. Keep permissions as small as possible and confirm consequential actions yourself.

Mini checklist

- ☐ I can explain MCP using an analogy (e.g. USB-C).
- ☐ I know the building blocks: client, server, tools.
- ☐ I grant read permissions before write permissions.
- ☐ I only use servers from trusted sources.

Exercise (5 min.)

Pick a use case from your everyday life (e.g. planning appointments, sorting invoices). Write down: what access does the AI really need for this (read/write, which folders/accounts)? What isn't needed? The result is your minimum-permissions list for a safe connection.

Next steps

- Extra exercises: `02_extra_exercises/17_extra_exercises`
- Test: Test E (Agentic systems)
- Related: 11 tools, 16 agents, 18 skills, 12 prompt injection

As of: October 2026